MTH401 Short Notes Mid term

Lec 1 to18

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Lecture#1

Ordinary Differential Equations

The term "**Ordinary Differential Equations**" also known as **ODE** is an equation that contains only one independent variable and one or more of its derivatives with respect to the variable. In other words, the ODE is represented as the relation having one independent variable x, the real dependent variable y, with some of its derivatives.

$y', y'', \ldots y^n, \ldots$ with respect to x.

Linear $y=mx+c$

Quadratic $ax^2+bx+c=0$

Cubic $ax^3+bx^2+cx+d=0$

Types

The ordinary differential equation is further classified into three types. They are:

- Autonomous ODE
- Linear ODE
- Non-linear ODE

Autonomous Ordinary Differential Equations

A differential equation which does not depend on the variable, say x is known as an autonomous differential equation.

Linear Ordinary Differential Equations

If differential equations can be written as the linear combinations of the derivatives of y, then they are called linear ordinary differential equations. These can be further classified into two types:

- Homogeneous linear differential equations
- Non-homogeneous linear differential equations

Non-linear Ordinary Differential Equations

If the differential equations cannot be written in the form of linear combinations of the derivatives of y, then it is known as a non-linear ordinary differential equation.

Example

$$y' = x^2 - 1$$
$$\frac{dy}{dx} = (x + y)^6$$
$$xy' = \sin x$$
$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + 5y = 0, y(0) = 0, y'(0) = 2$$
$$y'' = y' + xe^x$$

Lecture-2:

Elements of the Theory

Applicable to:

- Chemistry
- Physics
- Engineering
- Medicine
- Biology
- Anthropology

Differential Equation

involves an unknown function with one or more of its derivatives

Ordinary D.E.

a function where the unknown is dependent upon only one independent variable

Examples of Des

$$
\begin{aligned}
&\frac{dy}{dx}-5y &&=1\\
&(y-x)\,dx+4x\,dy &&=0\\
&\frac{d^2y}{dx^2}+5\left(\frac{dy}{dx}\right)^3-4y &&=e^x\\
&\frac{\partial u}{\partial y}+\frac{\partial v}{\partial x} &&=0\\
&x\frac{\partial u}{\partial x}+y\frac{\partial v}{\partial y} &&=u\\
&\frac{\partial^2 u}{\partial x^2}-\frac{\partial^2 u}{\partial t^2}+2\frac{\partial u}{\partial t} &&=0
\end{aligned}
$$

Specific Examples of ODE's

$\frac{du}{dt}=F(t)G(u),$	the growth equation;
$\frac{d^2\theta}{dt^2}+\frac{g}{l}\sin(\theta)=F(t),$	the pendulum equation;
$\frac{d^2y}{dt^2}+\varepsilon(y^2+1)\frac{dy}{dt}+y=0,$	the van der Pol equation;
$L\frac{d^2Q}{dt^2}+R\frac{dQ}{dt}+\frac{Q}{C}=E(t),$	the LCR oscillator equation;
$\frac{dp}{dt}=-2a(t)p+\frac{b(t)^2}{u(t)}p^2-v(t),$	a Riccati equation.

Ordinary Differential Equation

If an equation contains only ordinary derivatives of one or more dependent variables, w.r.t a single variable, then it is said to be an Ordinary Differential Equation (ODE). For example the differential equation

$$\frac{d^2y}{dx^2}+5\left(\frac{dy}{dx}\right)^3-4y=e^x$$

is an ordinary differential equation.

Partial Differential Equation

Similarly an equation that involves partial derivatives of one or more dependent variables w.r.t two or more independent variables is called a Partial Differential Equation (PDE).

For example the equation

$$a^2 \frac{\partial^4 u}{\partial x^4} + \frac{\partial^2 u}{\partial x^2} \qquad = 0$$

Results from ODE data „

- The solution of a general differential equation:
    - f(t, y, y', . . . , y(n)) = 0
- Is defined over some interval I having the following properties: „
    - y(t) and its first n derivatives exist for all t in I so that y(t) and its first n - 1 derivates must be continuous in I „
    - y(t) satisfies the differential equation for all t in I
- General Solution – all solutions to the differential equation can be represented in this form for all constants „
- Particular Solution – contains no arbitrary constants „
- Initial Condition
- Boundary Condition
- Initial Value Problem (IVP)
- Boundary Value Problem (BVP)

Properties of ODE's „

- Linear – if the nth-order differential equation can be written:

  $an(t)y(n) + an\text{-}1(t)y(n\text{-}1) + . . . + a1y' + a0(t)y = h(t)$
- Nonlinear – not linear $x3(y''')3\text{-}x2y(y'')2+3xy'+5y=ex$
- Superposition – allows us to decompose a problem into smaller, simpler parts and then combine them to find a solution to the original problem.

Implicit Solution

A relation G(x,y) is known as an implicit solution of a differential equation, if it defines one or more explicit solution on I.

Lecture 3

Separable Equations

Separable differential equations are a special type of differential equations where the variables involved can be separated to find the solution of the equation. Separable differential equations can be written in the form dy/dx = f(x) g(y), where x and y are the variables and are explicitly separated from each other.

Separable Differential Equation

$$\frac{dy}{dx} = f(x)\, g(y)$$

$$\Rightarrow \quad \frac{dy}{g(y)} = f(x)\, dx$$

$$\Rightarrow \quad \int \frac{1}{g(y)} dy = \int f(x)\, dx$$

Example

Find the particular solution of

$$\frac{dy}{dx} = \frac{y^2 - 1}{x}, \quad y(1) = 2$$

Solution:

1. By solving the equation

$$y^2 - 1 = 0$$

We obtain the constant solutions

$$y = \pm 1$$

2. Rewrite the equation as

$$\frac{dy}{y^2 - 1} = \frac{dx}{x}$$

Resolving into partial fractions and integrating, we obtain

$$\frac{1}{2}\int \left[\frac{1}{y-1} - \frac{1}{y+1}\right] dy = \int \frac{1}{x} dx$$

Integration of rational functions, we get

$$\frac{1}{2}\ln\frac{|y-1|}{|y+1|} = \ln|x| + C$$

3. The solutions to the given differential equation are

$$\begin{cases} \frac{1}{2}\ln\frac{|y-1|}{|y+1|} = \ln|x| + C \\ y = \pm 1 \end{cases}$$

4. Since the constant solutions do not satisfy the initial condition, we plug in the condition

$y = 2$ When $x = 1$ in the solution found in step 2 to find the value of C.

$$\frac{1}{2}\ln\left(\frac{1}{3}\right) = C$$

The above implicit solution can be rewritten in an explicit form as:

$$y = \frac{3 + x^2}{3 - x^2}$$

Example

Solve the differential equation

$$\frac{dy}{dt} = 1 + \frac{1}{y^2}$$

Solution:

1. We find roots of the equation to find constant solutions

$$1 + \frac{1}{y^2} = 0$$

No constant solutions exist because the equation has no real roots.

2. For non-constant solutions, we separate the variables and integrate

$$\int \frac{dy}{1+1/y^2} = \int dt$$

Since $\quad \frac{1}{1+1/y^2} = \frac{y^2}{y^2+1} = 1 - \frac{1}{y^2+1}$

Thus $\quad \int \frac{dy}{1+1/y^2} = y - \tan^{-1}(y)$

So that $\quad y - \tan^{-1}(y) = t + C$

It is **not easy** to find the **solution in an explicit form** i.e. y as a function of t.

3. Since $\exists$ no constant solutions, all solutions are given by the implicit equation found
in step 2.

Example

Solve

$$(1+x)dy - ydx = 0$$

Solution:

Dividing with $(1+x)y$, we can write the given equation as

$$\frac{dy}{dx} = \frac{y}{(1+x)}$$

1. The only constant solution is $y = 0$

2. For non-constant solution we separate the variables

$$\frac{dy}{y} = \frac{dx}{1+x}$$

Integrating both sides, we have

$$\int \frac{dy}{y} = \int \frac{dx}{1+x}$$

$$\ln|y| = \ln|1+x| + c_1$$

$$y = e^{\ln|1+x| + c_1} = e^{\ln|1+x|}.e^{c_1}$$

or $\quad y = |1+x|\, e^{c_1} == \pm e^{c_1}(1+x)$

$$y = C(1+x), \quad C = \pm e^{c_1}$$

If we use $\ln|c|$ instead of c_1 then the solution can be written as

$$\ln|y| = \ln|1+x| + \ln|c|$$

or $\quad \ln|y| = \ln|c(1+x)|$

So that $\quad y = c(1+x).$

3. The solutions to the given equation are

$$y = c(1+x)$$

$$y = 0$$

Lecture 4

Homogeneous Differential Equations

Homogeneous Differential Equations

A first order differential equation is homogeneous if it takes the form: $\frac{dy}{dx} = F\left(\frac{y}{x}\right)$,

Where $F\left(\frac{y}{x}\right)$, is a homogeneous function in this context homogeneous is used to mean a function of x and y that is left unchanged by multiplying both arguments by a constant, i.e.

$f(tx,ty) = t^n f(x,y)$ For some real number n, for any number t.

Example

Example 1

Determine whether the following functions are homogeneous

$$\begin{cases} f(x,y) = \dfrac{xy}{x^2+y^2} \\ g(x,y) = \ln\left(-3x^2y/(x^3+4xy^2)\right) \end{cases}$$

Solution:

The functions $f(x,y)$ is homogeneous because

$$f(tx,ty) = \frac{t^2xy}{t^2(x^2+y^2)} = \frac{xy}{x^2+y^2} = f(x,y)$$

Similarly, for the function $g(x,y)$ we see that

$$g(tx,ty) = \ln\left(\frac{-3t^3x^2y}{t^3(x^3+4xy^2)}\right) = \ln\left(\frac{-3x^2y}{x^3+4xy^2}\right) = g(x,y)$$

Therefore, the second function is also homogeneous.
Hence the differential equations

$$\begin{cases} \dfrac{dy}{dx} = f(x,y) \\ \dfrac{dy}{dx} = g(x,y) \end{cases}$$

Are homogeneous differential equations

Method of Solution:

To solve the homogeneous differential equation

$$\frac{dy}{dx} = f(x,y)$$

We use the substitution

$$v = \frac{y}{x}$$

If $f(x,y)$ is homogeneous of degree zero, then we have

$$f(x,y) = f(1,v) = F(v)$$

Since $y' = xv' + v$, the differential equation becomes

$$x\frac{dv}{dx} + v = f(1,v)$$

This is a separable equation. We solve and go back to old variable y through $y = xv$.

Example

Example 2 Solve the differential equation

$$\frac{dy}{dx}=\frac{-2x+5y}{2x+y}$$

Solution:

Step 1. It is easy to check that the function

$$f(x,y)=\frac{-2x+5y}{2x+y}$$

is a homogeneous function.

Step 2. To solve the differential equation we substitute

$$v=\frac{y}{x}$$

Step 3. Differentiating w.r.t x, we obtain

$$xv'+v=\frac{-2x+5xv}{2x+xv}=\frac{-2+5v}{2+v}$$

which gives

$$\frac{dv}{dx}=\frac{1}{x}\left(\frac{-2+5v}{2+v}-v\right)$$

This is a separable. At this stage please refer to the **Caution!**

Step 4. Solving by separation of variables all solutions are implicitly given by

$$-4\ln(|v-2|)+3\ln|v-1|=\ln(|x|)+C$$

Step 5. Going back to the function y through the substitution $y=vx$, we get

$$-4\ln|y-2x|+3\ln|y-x|=\ C$$

$$-4\ln\left|\frac{y-2x}{x}\right|+3\ln\left|\frac{y-x}{x}\right|=\ln|x|+c$$

$$\ln\left|\frac{y-2x}{x}\right|^{-4}+\ln\left|\frac{y-x}{x}\right|^{3}=\ln x+\ln c_1,\ c=\ln c_1$$

$$\ln\left|\frac{(y-2x)^{-4}}{x^{-4}}\right|+\ln\left|\frac{(y-x)^{3}}{x^{3}}\right|=\ln c_1x$$

$$\ln\left|\frac{(y-2x)^{-4}}{x^{-4}}.\frac{(y-x)^{3}}{x^{3}}\right|=\ln c_1x$$

$$\frac{(y-2x)^{-4}}{x^{-4}}.\frac{(y-x)^{3}}{x^{3}}=c_1x$$

$$x(y-2x)^{-4}(y-x)^{3}=c_1x$$

$$(y-2x)^{-4}(y-x)^{3}=c_1$$

Note that the implicit equation can be rewritten as

$$(y-x)^{3}=C_1(y-2x)^{4}$$

Differential Equations Reducible to Homogeneous Form

A differential equation of the form $\frac{dy}{dx}=\frac{ax+by+c}{a_1x+b_1y+c_1}$ where $\frac{a}{a_1}\neq\frac{b}{b_1}$ can be reduced to homogeneous form by taking new variable x and y such that x = X + h and y = Y + k, where h and k are constants to be so chosen as to make the given equation homogeneous. With the above substitutions, we get dx = dX and dy = dY, so that $\frac{dy}{dx}=\frac{dY}{dX}$.

The differential equation

$$\frac{dy}{dx}=\frac{a_1x+b_1y+c_1}{a_2x+b_2y+c_2}$$

is not homogenous. However, it can be reduced to a homogenous form as detailed below

Case 1: $\frac{a_1}{a_2}=\frac{b_1}{b_2}$

We use the substitution $z=a_1x+b_1y$ which reduces the equation to a separable equation in the variables x and z. Solving the resulting separable equation and replacing z with a_1x+b_1y, we obtain the solution of the given differential equation.

Case 2: $\frac{a_1}{a_2}\neq\frac{b_1}{b_2}$

In this case we substitute

$$x=X+h, \qquad y=Y+k$$

Where h and k are constants to be determined. Then the equation becomes

$$\frac{dY}{dX}=\frac{a_1X+b_1Y+a_1h+b_1k+c_1}{a_2X+b_2Y+a_2h+b_2k+c_2}$$

We choose h and k such that

$$\left.\begin{aligned}a_1h+b_1k+c_1=0\\a_2h+b_2k+c_2=0\end{aligned}\right\}$$

This reduces the equation to

$$\frac{dY}{dX}=\frac{a_1X+b_1Y}{a_2X+b_2Y}$$

Example

Solve the differential equation

$$\frac{dy}{dx}=-\frac{2x+3y-1}{2x+3y+2}$$

Solution:

Since $\frac{a_1}{a_2}=1=\frac{b_1}{b_2}$, we substitute $z=2x+3y$, so that

$$\frac{dy}{dx}=\frac{1}{3}\left(\frac{dz}{dx}-2\right)$$

Thus the equation becomes

$$\frac{1}{3}\left(\frac{dz}{dx}-2\right)=-\frac{z-1}{z+2}$$

i.e.

$$\frac{dz}{dx}=\frac{-z+7}{z+2}$$

This is a variable separable form, and can be written as

$$\left(\frac{z+2}{-z+7}\right)dz=dx$$

Integrating both sides we get

$$-z-9\ln(z-7)=x+A$$

Simplifying and replacing z with $2x+3y$, we obtain

$$-\ln(2x+3y-7)^9=3x+3y+A$$

or

$$(2x+3y-7)^{-9}=ce^{3(x+y)}, \qquad c=e^A$$

Lecture 5

Exact Differential Equations

Exact Differential Equation

A differential equation is an equation which contains one or more terms. It involves the derivative of one variable (dependent variable) with respect to the other variable (independent variable). The differential equation for a given function can be represented in a form: f(x) = dy/dx where "x" is an independent variable and "y" is a dependent variable. In this article, we are going to discuss what is an exact differential equation, standard form, integrating factor, and how to solve exact differential equation in detail with examples and solved problems.

Exact Differential Equation Definition

The equation P (x,y) dx + Q (x,y) dy=0 is an exact differential equation if there exists a function f of two variables x and y having continuous partial derivatives such that the exact differential equation definition is separated as follows

$$u_x(x, y) = p(x, y) \text{ and } u_y(x, y) = Q(x, y);$$

Therefore, the general solution of the equation is u(x, y) = C. Where "C" is an arbitrary constant.

Testing for Exactness

Assume the functions P(x, y) and Q(x, y) having the continuous partial derivatives in a particular domain D, and the differential equation is exact if and only if it satisfies the condition $\frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y}$.

Exact Differential Equation Integrating Factor

If the differential equation P (x, y) dx + Q (x, y) dy = 0 is not exact, it is possible to make it exact by multiplying using a relevant factor u(x, y) which is known as integrating factor for the given differential equation.

Method of Solution:

If the given equation is exact then the solution procedure consists of the following steps:

Step 1. Check that the equation is exact by verifying the condition $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

Step 2. Write down the system $\frac{\partial F}{\partial x} = M(x,y)$, $\frac{\partial F}{\partial y} = N(x,y)$

Step 3. Integrate either the 1st equation *w. r. to* $\boldsymbol{x}$ or 2nd *w. r. to* $\boldsymbol{y}$. If we choose the 1st equation then

$$F(x,y) = \int M(x,y)dx + \theta(y)$$

The function $\theta(y)$ is an arbitrary function of y, integration w.r.to x; y being constant.

Step 4. Use second equation in step 2 and the equation in step 3 to find $\theta'(y)$.

$$\frac{\partial F}{\partial y} = \frac{\partial}{\partial y}\left(\int M(x,y)dx\right) + \theta'(y) = N(x,y)$$

$$\theta'(y) = N(x,y) - \frac{\partial}{\partial y}\int M(x,y)dx$$

Step 5. Integrate to find $\theta(y)$ and write down the function $F(x,y)$;

Step 6. All the solutions are given by the implicit equation

$$F(x,y) = C$$

Step 7. If you are given an IVP, plug in the initial condition to find the constant C.

Example

Solve the initial value problem

$$(\cos x \sin x - xy^2)dx + y(1-x^2)dy = 0,\ y(0) = 2$$

Solution:

Since
$$\begin{cases} M(x,y) = \cos x.\sin x - x\,y^2 \\ N(x,y) = \quad y(1-x^2) \end{cases}$$

and
$$\frac{\partial M}{\partial y} = -2xy = \frac{\partial N}{\partial x}$$

Therefore the equation is exact and $\exists$ a function $f(x,y)$ such that

$$\frac{\partial f}{\partial x} = \cos x\,.\,sin\,x - x\,y^2 \quad \text{and} \quad \frac{\partial f}{\partial y} = y(1-x^2)$$

Now integrating 2nd of these equations w.r.t. 'y' keeping 'x' constant, we obtain

$$f(x,y) = \frac{y^2}{2}(1-x^2) + h(x)$$

Differentiate w.r.t. 'x' and equate the result to $M(x,y)$

$$\frac{\partial f}{\partial x} = -xy^2 + h'(x) = \cos x \sin x - xy^2$$

The last equation implies that.

$$h'(x) = \cos x \sin x$$

Integrating w.r.to x, we obtain

$$h(x) = -\int(\cos x)(-\sin x)dx = -\frac{1}{2}\cos^2 x$$

Thus a one parameter family solutions of the given differential equation is

$$\boxed{\frac{y^2}{2}\left(1-x^2\right)-\frac{1}{2}\cos^2 x=c_1}$$

or

$$\boxed{y^2\left(1-x^2\right)-\cos^2 x=c}.$$

where $2c_1$ has been replaced by C. The initial condition $y=2$ when $x=0$ demand, that $4(1)-\cos^2(0)=c$ so that $c=3$. Thus the solution of the initial value problem is

$$\boxed{y^2\left(1-x^2\right)-\cos^2 x=3}$$

Lecture – 6

Integrating Factor Technique

Integrating Factor Technique

integrating factor is a function used to solve differential equations. It is a function in which an ordinary differential equation can be multiplied to make the function integrable. It is usually applied to solve ordinary differential equations. Also, we can use this factor within multivariable calculus. When multiplied by an integrating factor, an inaccurate differential is made into an accurate differential (which can be later integrated to give a scalar field). It has a major application in thermodynamics where the temperature becomes the integrating factor that makes entropy an exact differential.

Integrating Factor Method

Integrating factor is defined as the function which is selected in order to solve the given differential equation. It is most commonly used in ordinary linear differential equations of the first order.

When the given differential equation is of the form; dy/dx + P(x) y = Q(x) where P and Q are functions involving x only.

$$\frac{dy}{dx} + \frac{3y}{x} = \frac{e^x}{x^3} \qquad \text{or} \qquad \frac{dy}{dx} - \frac{3y}{x+1} = (x+1)^4.$$

We can solve these differential equations using the technique of an integrating factor. Where P(x) (the function of x) is a multiple of y and μ denotes integrating factor.

Solving First-Order Differential Equation Using Integrating Factor

Below are the steps to solve the first-order differential equation using the integrating factor.

- Compare the given equation with differential equation form and find the value of P(x).
- Calculate the integrating factor μ.
- Multiply the differential equation with integrating factor on both sides in such a way; μ dy/dx + μP(x)y = μQ(x)
- In this way, on the left-hand side, we obtain a particular differential form. I.e d/dx(μ y) = μQ(x)
- In the end, we shall integrate this expression and get the required solution to the given equation: μ y = ∫μQ(x)dx+C

Solving Second Order Differential Equation Using Integrating Factor

The second-order differential equation can be solved using the integrating factor method.

Let the given differential equation be,

y” + P(x) y’ = Q(x)

The second-order equation of the above form can only be solved by using the integrating factor.

- Substitute y’ = u; so that the equation becomes similar to the first-order equation as shown: $u' + P(x)\, u = Q(x)$
- Now, this equation can be solved by integrating factor technique as described in the section above for first-order equations and we reach the equation: $\mu\, u=\int \mu Q(x)dx+C$
- Find the value of u from this equation.

Since u = y’, hence to find the value of y, integrate the equation. In this way, we get the required solution.

Integrating Factor Example

Example

Solve the differential equation using the integrating factor: $(dy/dx) – (3y/x+1) = (x+1)^4$

Solution:

Given: $(dy/dx) – (3y/x+1) = (x+1)^4$

First, find the integrating factor:

$\mu = e^{\int p(x)\, dx}$

$\mu = e^{\int(-3/x+1)\, dx}$

$\int(-3/ x+1)dx = -3 \ln (x+1) = \ln (x+1)^{-3}$

Hence, we get

$\mu =e^{\ln (x+1)-3}$

$\mu = 1/ (x+1)^3$

Now, multiply the integrating factor on both the sides of the given differential eqaution:

$[1/ (x+1)^3] [Dy/dx] – [3y/ ((x+1)^4)] = (x+1)$

Integrate both the sides, we get:

$[y/(x+1)^3] = [(1/2) x^2+x+c]$

Here, c is a constant

Therefore, the general solution of the given differential eqaution is

y = [(x+1)^3] [(1/2)x^2+x+c].

Example

Example 4

Solve $$\left(x^2y-2xy^2\right)dx-\left(x^3-3x^2y\right)dy=0$$

Solution: Comparing with

$$Mdx+Ndy=0$$

we see that

$$M=x^2y-2xy^2 \quad \text{and} \quad \mathrm{N}=-(x^3-3x^2y)$$

Since both M and N are homogeneous. Therefore, the given equation is homogeneous.
Now

$$xM+yN=x^3y-2x^2y^2-x^3y+3x^2y^2=x^2y^2\neq 0$$

Hence, the factor u is given by

$$u=\frac{1}{x^2y^2} \qquad \because u=\frac{1}{xM+yN}$$

Multiplying the given equation with the integrating factor u, we obtain.

$$\left(\frac{1}{y}-\frac{2}{x}\right)dx-\left(\frac{x}{y^2}-\frac{3}{y}\right)dy=0$$

Now

$$M=\frac{1}{y}-\frac{2}{x} \quad \text{and} \quad \mathrm{N}=\frac{-x}{y^2}+\frac{3}{y}$$

and therefore

$$\frac{\partial M}{\partial y}=-\frac{1}{y^2}=\frac{\partial N}{\partial x}$$

Therefore, the new equation is exact and solution of this new equation is given by

$$\frac{x}{y}-2\ln|x|+3\ln|y|=C$$

Lecture 7

First Order Linear Equations

First Order Linear Equations

A **first order linear differential equation** is a differential equation of the form $y'+p(x)y=q(x)$. The left-hand side of this equation looks almost like the result of using the product rule, so we solve the equation by multiplying through by a factor that will make the left-hand side exactly the result of a product rule, and then integrating. This factor is called an integrating factor.

First order linear differential equations are the only differential equations that can be solved even with variable coefficients almost every other kind of equation that can be solved explicitly requires the coefficients to be constant, making these one of the broadest classes of differential equations that can be solved.

Integrating Factors

An integrating factor will be a function $f(x)$ such that multiplying it to both sides of the equation,

$$f(x)\left[y' + p(x)y\right] = f(x)q(x)$$

$$f(x)y' + f(x)p(x)y = f(x)q(x),$$

gives an expression on the left-hand side that can be integrated with the anti-product rule. In other words, it is the function that satisfies:

$$f(x)y' + f(x)p(x)y = f(x)y' + f'(x)y$$

$$\int \left[f(x)y' + f(x)p(x)y\right]\,dx = f(x)y + C$$

This means the function f should satisfy $f'(x) = f(x)\,p(x)$, which is a separable differential equation. This can be solved as

$$\frac{f'(x)}{f(x)}\,dx = p(x)\,dx \implies \int \frac{f'(x)}{f(x)}\,dx = \int p(x)\,dx$$

$$\implies \ln|f(x)| = \int p(x)\,dx$$

$$\implies f(x) = e^{\int p(x)\,dx}.$$

Method of solution:

1. Identify that the equation is 1st order linear equation. Rewrite it in the form

$$\frac{dy}{dx} + p(x)y = q(x)$$

if the equation is not already in this form.

2. Find the integrating factor

$$u(x) = e^{\int p(x)dx}$$

3. Write down the general solution

$$y = \frac{\int u(x)q(x)dx + C}{u(x)}$$

4. If you are given an IVP, use the initial condition to find the constant *C*.

5. Plug in the calculated value to write the particular solution of the problem.

Example

Solve the initial value problem

$$y' + \tan(x)y = \cos^2(x), \qquad y(0) = 2$$

Solution:

1.The equation is already in the standard form

$$\frac{dy}{dx} + p(x)y = q(x)$$

with

$$\begin{cases} p(x) = \tan x \\ q(x) = \cos^2 x \end{cases}$$

2. Since

$$\int \tan x\,dx = -\ln\cos x = \ln\sec x$$

Therefore, the integrating factor is given by

$$u(x) = e^{\int \tan x\,dx} = \sec x$$

3. Further, because

$$\int \sec x\cos^2 x\,dx = \int \cos x\,dx = \sin x$$

So that the general solution is given by

$$y = \frac{\sin x + C}{\sec x} = (\sin x + C)\cos x$$

4. We use the initial condition $y(0) = 2$ to find the value of the constant C

$$y(0) = C = 2$$

5. Therefore the solution of the initial value problem is

$$y = (\sin x + 2)\cos x$$

Lecture 8

Bernoulli Equations

Bernoulli Equations

A differential equation that can be written in the form $\frac{dy}{dx} + p(x)y = q(x)y^n$ is called Bernoulli equation.

- Method of solution: For n = 1,0 the equation reduces to 1st order linear DE and can be solved
- Accordingly. For n ≠ 1,0 we divide the equation with y^n to write it in the form

$$y^{-n}\frac{dy}{dx} + p(x)y^{1-n} = q(x)$$

and then put

$$v = y^{1-n}$$

Differentiating w.r.t. 'x', we obtain

$$v' = (1-n)y^{-n}y'$$

Therefore the equation becomes

$$\frac{dv}{dx} + (1-n)p(x)v = (1-n)q(x)$$

This is a linear equation satisfied by v. Once it is solved, you will obtain the function

$$y = v^{\frac{1}{(1-n)}}$$

If n > 1, then we add the solution y = 0 to the solutions found the above technique.

Example

Solve $$\frac{dy}{dx}+\frac{1}{x}y=xy^2$$

Solution: In the given equation we identify $P(x)=\frac{1}{x},\ q(x)=x$ and $n=2$.

Thus the substitution $w=y^{-1}$ gives

$$\frac{dw}{dx}-\frac{1}{x}w=-x.$$

The integrating factor for this linear equation is

$$e^{-\int\frac{dx}{x}}=e^{-\ln|x|}=e^{\ln|x|^{-1}}=x^{-1}$$

Hence $$\frac{d}{dx}\left[x^{-1}w\right]=-1.$$

Integrating this latter form, we get

$$x^{-1}w=-x+c \text{ or } w=-x^2+cx.$$

Since $w=y^{-1}$, we obtain $y=\frac{1}{w}$ or

$$y=\frac{1}{-x^2+cx}$$

For n > 0 the trivial solution y = 0 is a solution of the given equation. In this example, y = 0 is a singular solution of the given equation.

SUBSTITUTIONS ‰

- Sometimes a differential equation can be transformed by means of a substitution into a form that could then be solved by one of the standard methods i.e. Methods used to solve separable, homogeneous, exact, linear, and Bernoulli's differential equation. ‰
- An equation may look different from any of those that we have studied in the previous lectures, but through a sensible change of variables perhaps an apparently difficult problem may be readily solved. ‰
- Although no firm rules can be given on the basis of which these substitution could be selected, a working axiom might be: Try something! It sometimes pays to be clever.

Example

$$x\frac{dy}{dx}-y=\frac{x^3}{y}e^{y/x}$$

Solution:

If we let

$$u=\frac{y}{x}$$

Then the given differential equation can be simplified to

$$ue^{-u}du=dx$$

Integrating both sides, we have

$$\int ue^{-u}du=\int dx$$

Using the integration by parts on LHS, we have

$$-ue^{-u}-e^{-u}=x+c$$

or

$$u+1=(c_1-x)e^{u} \text{ Where } c_1=-c$$

We then re-substitute

$$u=\frac{y}{x}$$

and simplify to obtain

$$y+x=x(c_1-x)e^{y/x}$$

Lecture-09 Example

Very important 5 Examples

$$\frac{dy}{dx}=\frac{(2\sqrt{xy}-y)}{x}$$

Solution: $\frac{dy}{dx}=\frac{(2\sqrt{xy}-y)}{x}$

put $y=wx$

$$w+x\frac{dw}{dx}=\frac{(2\sqrt{xwx}-xw)}{x}$$

$$w+x\frac{dw}{dx}=2\sqrt{w}-w$$

$$x\frac{dw}{dx}=2\sqrt{w}-2w$$

$$\frac{dw}{2(\sqrt{w}-w)}=\frac{dx}{x}$$

$$\int\frac{dw}{2(\sqrt{w}-w)}=\int\frac{dx}{x}$$

$$\int\frac{dw}{2\sqrt{w}(1-\sqrt{w})}=\int\frac{dx}{x}$$

put $\sqrt{w}=t$

We get $\int\frac{1}{1-t}dt=\int\frac{dx}{x}$

$$-\ln|1-t|=\ln|x|+\ln|c|$$

$$-\ln|1-t|=\ln|xc|$$

$$(1-t)^{-1}=xc$$

$$(1-\sqrt{w})^{-1}=xc$$

$$(1-\sqrt{y/x})^{-1}=xc$$

Example $\frac{dy}{dx}=\frac{2xye^{(x/y)^2}}{y^2+y^2e^{(x/y)^2}+2x^2e^{(x/y)^2}}$

Solution: $\frac{dx}{dy}=\frac{y^2+y^2e^{(x/y)^2}+2x^2e^{(x/y)^2}}{2xye^{(x/y)^2}}$

put $x/y=w$

Aftersubsitution

$$y\frac{dw}{dy}=\frac{1+e^{w^2}}{2we^{w^2}}$$

$$\frac{dy}{y}=\frac{2we^{w^2}}{1+e^{w^2}}dw$$

Integrating

$$\ln|y|=\ln|1+e^{w^2}|+\ln c$$

$$\ln|y|=\ln|c(1+e^{w^2})|$$

$$y=c(1+e^{(x/y)^2})$$

Example $\frac{dy}{dx}+\frac{y}{x\ln x}=\frac{3x^2}{\ln x}$

Solution: $\frac{dy}{dx}+\frac{y}{x\ln x}=\frac{3x^2}{\ln x}$

$$\frac{dy}{dx}+\frac{1}{x\ln x}y=\frac{3x^2}{\ln x}$$

$p(x)=\frac{1}{x\ln x}$ and $q(x)=\frac{3x^2}{\ln x}$

$$I.F=\exp(\int\frac{1}{x\ln x}dx)=\ln x$$

Multiply both side by lnx

$$\ln x\frac{dy}{dx}+\frac{1}{x}y=3x^2$$

$$\frac{d}{dx}(y\ln x)=3x^2$$

Integrate

$$y\ln x=\frac{3x^3}{3}+c$$

Example

$$x\cos x\frac{dy}{dx}+y(x\sin x+\cos x)=1$$

Solution: $x\cos x\frac{dy}{dx}+y(x\sin x+\cos x)=1$

$$\frac{dy}{dx}+y\left[\frac{x\sin x+\cos x}{x\cos x}\right]=\frac{1}{x\cos x}$$

$$\frac{dy}{dx}+y\left[\tan x+1/x\right]=\frac{1}{x\cos x}$$

$$I.F=\exp(\int(\tan x+1/x)dx)=x\sec x$$

$$x\sec x\frac{dy}{dx}+yx\sec x\left[\tan x+1/x\right]=\frac{x\sec x}{x\cos x}$$

$$x\sec x\frac{dy}{dx}+y\left[x\sec x\tan x+\sec x\right]=\sec^2 x$$

$$\frac{d}{dx}\left[xy\sec x\right]=\sec^2 x$$

$$xy\sec x=\tan x+c$$

Example $2x\csc 2y\frac{dy}{dx}=2x-\ln\tan y$

Solution: $2x\csc 2y\frac{dy}{dx}=2x-\ln\tan y$

put $\ln\tan y=u$

$$\frac{dy}{dx}=\sin y\cos y\frac{du}{dx}$$

$$\frac{2x\sin y\cos y}{2\sin y\cos y}\frac{du}{dx}=2x-u$$

$$x\frac{du}{dx}=2x-u$$

$$\frac{du}{dx}+\frac{1}{x}u=2$$

$$I.F=\exp(\int 1/x\,dx)=x$$

$$x\frac{du}{dx}+u=2x$$

$$\frac{d}{dx}(xu)=2x$$

$$xu=x^2+c$$

$$u=x+cx^{-1}$$

$$\ln\tan y=x+cx^{-1}$$

Lecture-10

Applications of First Order Differential Equations

Model for the phenomenon

Translate a physical phenomenon in terms of mathematics; we strive for a set of equations that describe the system adequately. This set of equations is called a Model for the phenomenon.

A **first-order differential equation** is defined by an equation: **$dy/dx = f(x,y)$** of two variables x and y with its function f(x,y) defined on a region in the xy-plane. It has only the first derivative dy/dx so that the equation is of the first order and no higher-order derivatives exist. The differential equation in first-order can also be written as;

$$y' = f(x,y) \quad \text{or}$$

$$(d/dx)\, y = f(x,y)$$

The differential equation is generally used to express a relation between the function and its derivatives. In Physics and chemistry, it is used as a technique for determining the functions over its domain if we know the functions and some of the derivatives.

First Order Linear Differential Equation

If the function f is a linear expression in y, then the first-order differential equation y' = f (x, y) is a linear equation. That is, the equation is linear and the function f takes the form

$$f(x,y) = p(x)y + q(x)$$

Since the linear equation is $y = mx+b$

Where p and q are continuous functions on some interval I. Differential equations that are not linear are called nonlinear equations.

Consider the first-order differential equation $y' = f(x,y)$, is a linear equation and it can be written in the form

- **$y' + a(x)y = f(x)$**

Where a(x) and f(x) are continuous functions of x

The alternate method to represent the first-order linear equation in a reduced form is

$$(dy/dx) + P(x)y = Q(x)$$

Where P(x) and Q(x) are the functions of x which are the continuous functions. If P(x) or Q(x) is equal to zero, the differential equation is reduced to the variable separable form. It is easy to solve when the differential equations are in variable separable form.

First Order Differential Equations Solutions

Usually, there are two methods considered to solve the linear differential equation of first order.

1. Using Integrating Factor
2. Method of variation of constant

Let us discuss each method one by one to get the solutions for differential equations of the first order.

Integrating Factor

If a linear differential equation is written in the standard form:

$$y' + a(x)y = 0$$

Then, the integrating factor is defined by the formula

$$u(x) = \exp\left(\int a(x)dx\right)$$

Multiplying the integrating factor u(x) on the left side of the equation that converts the left side into the derivative of the product y(x)u(x).

The general solution of the differential equation is expressed as follows:

$$y = \frac{\int u(x)f(x)dx + C}{u(x)}$$

Where C is an arbitrary constant.

Method of Variation of a Constant

This method is similar to the integrating factor method. Finding the general solution of the homogeneous equation is the first necessary step.

$Y' + a(x)\, y = 0$

Example

Example

Find the Orthogonal Trajectory to the family of circles

$$x^2 + y^2 = 2Cx$$

Solution:

1. We differentiate the given equation to find the DE satisfied by the circles.

$$y\frac{dy}{dx} + x = C, \qquad C = \frac{x^2 + y^2}{2x}$$

2. The explicit differential equation associated to the family of circles is

$$\frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$$

3. Hence the differential equation for the orthogonal family is

$$\frac{dy}{dx} = \frac{2xy}{x^2 - y^2}$$

4. This DE is a homogeneous, to solve this equation we substitute $v = y/x$

or equivalently $y = vx$. Then we have

$$\frac{dy}{dx} = x\frac{dv}{dx} + v \text{ and } \frac{2xy}{x^2 - y^2} = \frac{2v}{1 - v^2}$$

Therefore the homogeneous differential equation in step 3 becomes

$$x\frac{dv}{dx} + v = \frac{2v}{1 - v^2}$$

Algebraic manipulations reduce this equation to the separable form:

$$\frac{dv}{dx} = \frac{1}{x}\left\{\frac{v + v^3}{1 - v^2}\right\}$$

The constant solutions are given by

$$v + v^3 = 0 \Rightarrow v(1 + v^2) = 0$$

The only constant solution is $v = 0$.

To find the non-constant solutions we separate the variables

$$\frac{1 - v^2}{v + v^3}dv = \frac{1}{x}dx$$

Integrate

$$\int \frac{1 - v^2}{v + v^3} dv = \int \frac{1}{x} dx$$

Resolving into partial fractions the integrand on LHS, we obtain

$$\frac{1 - v^2}{v + v^3} = \frac{1 - v^2}{v(1 + v^2)} = \frac{1}{v} - \frac{2v}{1 + v^2}$$

Hence we have

$$\int \frac{1 - v^2}{v + v^3} dv = \int \left\{\frac{1}{v} - \frac{2v}{1 + v^2}\right\} dv = \ln|v| - \ln[v^2 + 1]$$

Hence the solution of the separable equation becomes

$$\ln|v| - \ln[v^2 + 1] = \ln|x| + \ln C$$

which is equivalent to

$$\frac{v}{v^2 + 1} = Cx$$

where $C \neq 0$. Hence all the solutions are

$$\begin{cases} v &= 0 \\ \dfrac{v}{v^2 + 1} &= Cx \end{cases}$$

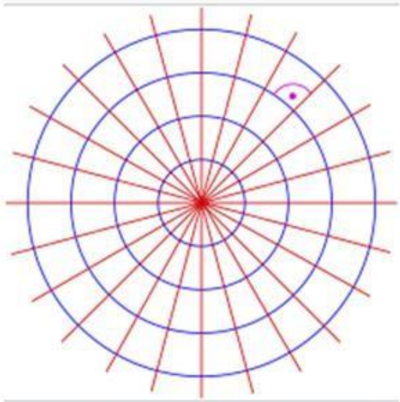
We go back to y to get $y = 0$ and $\dfrac{y}{y^2 + x^2} = C$ which is equivalent to

$$\begin{cases} y &= 0 \\ x^2 + y^2 &= my \end{cases}$$

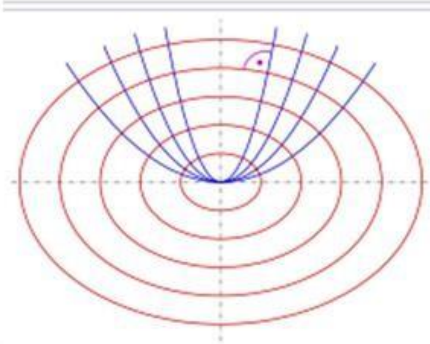
Orthogonal trajectory

Orthogonal trajectory is a curve, which intersects any curve of a given pencil of (planar) curves *orthogonally*.

For example, the orthogonal trajectories of a pencil of *concentric circles* are the lines through their common center



Concentric circles with orthogonal trajectories (1. example)



Parabolas with orthogonal trajectories (2. example)

Isogonic trajectory.

Orthogonal trajectories are used in mathematics for example as curved coordinate systems (i.e. elliptic coordinates) or appear in physics as electric fields and their equipotential curves.

If the trajectory intersects the given curves by an arbitrary (but fixed) angle, one gets an **isogonic trajectory**.

Population dynamics

Population dynamics is the portion of ecology that deals with the variation in time and space of population size and density for one or more species (Begon et al. 1990). In practice investigations and theory on population dynamics can be viewed as having two broad components:

first, quantitative descriptions of the changes in population number and form of population growth or decline for a particular organism, and second, investigations of the forces and biological and physical processes causing those changes. The first of these components involves descriptive data that are useful for quantifying trends, and with appropriate statistical treatment, for forecasting future trends.

Lecture-11

Radioactive Decay

Radioactive Decay

Radioactive decay occurs in certain isotopes, which are variations of an element with different numbers of neutrons in their nuclei. These isotopes are said to be radioactive because they are unstable and tend to transform into more stable isotopes or different elements altogether.

The rate at which radioactive decay happens is characterized by a constant called the decay constant, usually denoted by λ (lambda). The decay constant determines the probability of an individual nucleus undergoing decay per unit time. It is related to the half-life of the isotope, which is the time it takes for half of the initial quantity of the substance to decay.

Mathematically, radioactive decay is often modeled using the exponential decay equation:
$N(t) = N_0 * e^{(-\lambda t)}$
In this equation:

- $N(t)$ represents the quantity of the radioactive substance at time t.
- N_0 is the initial quantity of the substance at t = 0.
- e is the mathematical constant approximately equal to 2.71828.
- λ is the decay constant.

Example:

A radioactive isotope has a half-life of 16 days. We have 30 g at the end of 30 days. How much radioisotope was initially present?
Solution:
Let tA)(be the amount present at time t and A_0 the initial amount of the isotope. Then we have to solve the initial value problem.

$$\frac{dA}{dt} = kA, \quad A(30) = 30$$

We know that the solution of the IVP is given by

$$A(t) = A_0 e^{kt}$$

If T the half-life then the constant is given k by

$$kT = -\ln 2 \quad \text{or} \quad k = -\frac{\ln 2}{T} = -\frac{\ln 2}{16}$$

Now using the condition $A(30) = 30$, we have

$$30 = A_0 e^{30k}$$

So that the initial amount is given by

$$A_0 = 30e^{-30k} = 30e^{\frac{30\ln 2}{16}} = 110.04\,\text{g}$$

Newton's Law of Cooling

From experimental observations it is known that the temperature T (t) of an object changes at a rate proportional to the difference between the temperature in the body and the temperature Tm of the surrounding environment. This is what is known as Newton's law of cooling.

If initial temperature of the cooling body is T0then we obtain the initial value problem

$$\frac{dT}{dt} = k(T - T_m), \quad T(0) = T_0$$

where k is constant of proportionality. The differential equation in the problem is linear as well as separable.

Separating the variables and integrating we obtain

$$\int \frac{dT}{T - T_m} = \int k\,dt$$

This means that

$$\ln|T - T_m| = kt + C$$

$$T - T_m = e^{kt+C}$$

$$T(t) = T_m + C_1 e^{kt} \quad \text{where} \quad C_1 = e^C$$

Now applying the initial condition $T(0) = T_0$, we see that $C_1 = T_0 - T_m$. Thus the solution of the initial value problem is given by

$$T(t) = T_m + (T_0 - T_m)e^{kt}$$

Hence, If temperatures at times t_1 and t_2 are known then we have

$$T(t_1) - T_m = (T_0 - T_m)e^{kt_1}, \quad T(t_2) - T_m = (T_0 - T_m)e^{kt_2}$$

So that we can write

$$\frac{T(t_1) - T_m}{T(t_2) - T_m} = e^{k(t_1 - t_2)}$$

This equation provides the value of k if the interval of time '$t_1 - t_2$' is known and vice-versa.

Carbon Dating ‰

- The isotope C–14 is produced in the atmosphere by the action of cosmic radiation on nitrogen. ‰
- The ratio of C-14 to ordinary carbon in the atmosphere appears to be constant. ‰
- The proportionate amount of the isotope in all living organisms is same as that in the atmosphere. ‰
- When an organism dies, the absorption of C −14 by breathing or eating ceases. ‰

- Thus comparison of the proportionate amount of C −14 present, say, in a fossil with constant ratio found in the atmosphere provides a reasonable estimate of its age. ‰
- The method has been used to date wooden furniture in Egyptian tombs. ‰
- Since the method is based on the knowledge of half-life of the radioactive C −14 (5600 years approximately), the initial value problem discussed in the radioactivity model governs this analysis.

Example:

A fossilized bone is found to contain 1000/1 of the original amount of C–14. Determine the age of the fissile.

Solution:

Let A(t) be the amount present at any time t and A_0 the original amount of C–14. Therefore, the process is governed by the initial value problem.

$$\frac{dA}{dt} = kA, \quad A(0) = A_0$$

We know that the solution of the problem is

$$\boxed{A(t) = A_0 e^{kt}}$$

Since the half life of the carbon isotope is 5600 years. Therefore,

$$\boxed{A(5600) = \frac{A_0}{2}}$$

So that

$$\boxed{\frac{A_0}{2} = A_0 e^{5600k} \quad \text{or} \quad 5600k = -\ln 2}$$

$$\boxed{k = -0.00012378}$$

Hence

$$\boxed{A(t) = A_0 e^{-(0.00012378)t}}$$

If t denotes the time when fossilized bone was found then $\boxed{A(t) = \frac{A_0}{1000}}$

$$\boxed{\frac{A_0}{1000} = A_0 e^{-(0.00012378)t} \Rightarrow -0.00012378\,t = -\ln 1000}$$

Therefore

$$\boxed{t = \frac{\ln 1000}{0.00012378} = 55{,}800 \text{ years}}$$

Lecture-12

Applications of Nonlinear Equations

Applications of Nonlinear Equations

As we know that the solution of the exponential model for the population growth is

$$\boxed{P(t) = P_0 e^{kt}}$$

P_0 being the initial population. From this solution we conclude that

(a) If $k > 0$ the population grows and expand to infinity i.e. $\boxed{\lim_{t \to \infty} P(t) = +\infty}$

(b) If $k < 0$ the population will shrink to approach 0, which means extinction.

Note that:

(1) The prediction in the first case ($k > 0$) differs substantially from what is actually observed, population growth is eventually limited by some factor**!**

(2) Detrimental effects on the environment such as pollution and excessive and competitive demands for food and fuel etc. can have inhibitive effects on the population growth.

Logistic equation:

Another model was proposed to remedy this flaw in the exponential model. This is called the **logistic model** (also called **Verhulst-Pearl model**).

Suppose that $a > 0$ is constant average rate of birth and that the death rate is proportional to the population $P(t)$ at any time t . Thus if $\frac{1}{P}\frac{dP}{dt}$ is the rate of growth per individual then

$$\boxed{\frac{1}{P}\frac{dP}{dt} = a - bP \quad \textbf{or} \quad \frac{dP}{dt} = P(a - bP)}$$

Where b is constant of proportionality. The term $\boxed{-bP^2,\ b > 0}$ can be interpreted as inhibition term. When $b = 0$, the equation reduces to the one in exponential model.

Solution to the logistic equation is also very important in ecological, sociological and even in managerial sciences. The logistic equation

$$\boxed{\frac{dP}{dt} = P(a - bP)}$$

can be easily identified as a **nonlinear** equation that is separable. The constant solutions of the equation are given by

$$P(a-bP)=0$$

$$\Rightarrow \quad \mathrm{P}=0 \quad \text{and} \quad P=\frac{a}{b}$$

For non-constant solutions we separate the variables

$$\frac{dP}{P(a-bP)}=dt$$

Resolving into partial fractions we have

$$\left[\frac{1/a}{P}+\frac{b/a}{a-bP}\right]dP=dt$$

Integrating

$$\frac{1}{a}\ln|P|-\frac{1}{a}\ln|a-bP|=t+C$$

$$\ln\left|\frac{P}{a-bP}\right|=at+aC$$

or

$$\frac{P}{a-bP}=C_1e^{at} \quad \text{where} \quad C_1=e^{aC}$$

Easy algebraic manipulations give

$$P(t)=\frac{aC_1e^{at}}{1+bC_1e^{at}}=\frac{aC_1}{bC_1+e^{-at}}$$

Here C_1 is an arbitrary constant. If we are given the initial condition $P(0)=P_0$, $P_0\neq\frac{a}{b}$ we obtain $C_1=\frac{P_0}{a-bP_0}$. Substituting this value in the last equation and simplifying, we obtain

$$P(t) = \frac{aP_0}{bP_0 + (a - bP_0)e^{-at}}$$

Clearly $\lim_{t \to \infty} P(t) = \frac{aP_0}{bP_0} = \frac{a}{b}$, **limited growth**

Note that $P = \frac{a}{b}$ is a **singular solution** of the logistic equation.

Special Cases of Logistic Equation:

1. Epidemic Spread

   Suppose that one person infected from a contagious disease is introduced in a fixed population of n people. The natural assumption is that the rate $\frac{dx}{dt}$ of spread of disease is proportional to the number $x(t)$ of the infected people and number $y(t)$ of people not infected people. Then

$$\frac{dx}{dt} = kxy$$

Since $x + y = n + 1$

Therefore, we have the following initial value problem

$$\frac{dx}{dt} = kx(n + 1 - x), \quad x(0) = 1$$

The last equation is a **special case of the logistic equation** and has also been used for the **spread of information** and the **impact of advertising** in centers of population.

2. A Modification of LE:

   A modification of the nonlinear logistic differential equation is the following

$$\frac{dP}{dt} = P(a - b\ln P)$$

has been used in the studies of solid tumors, in actuarial predictions, and in the growth of revenue from the sale of a commercial product in addition to growth or decline of population.

Chemical reactions

In a first order chemical reaction, the molecules of a substance A decompose into smaller molecules. This decomposition takes place at a rate proportional to the amount of the first substance that has not undergone conversion. The disintegration of a radioactive substance is an example of the first order reaction. If X is the remaining amount of the substance A at any time t then

$$\frac{dX}{dt} = kX$$

$k < 0$ Because X is decreasing.

In a 2nd order reaction two chemicals A and B react to form another chemical C at a rate proportional to the product of the remaining concentrations of the two chemicals.

If X denotes the amount of the chemical C that has formed at time t. Then the instantaneous amounts of the first two chemicals A and B not converted to the chemical C are $\alpha - X$ and $\beta - X$, respectively. Hence the rate of formation of chemical C is given by

$$\frac{dX}{dt} = k(\alpha - X)(\beta - X)$$

where k is constant of proportionality.

Miscellaneous Applications

- The velocity v of a falling mass m, subjected to air resistance proportional to instantaneous velocity, is given by the differential equation

$$m\frac{dv}{dx} = mg - kv$$

Here $k > 0$ is constant of proportionality.

- The rate at which a drug disseminates into bloodstream is governed by the differential equation

$$\frac{dx}{dt} = A - Bx$$

Here A, B are positive constants and $x(t)$ describes the concentration of drug in the bloodstream at any time t.

- The rate of memorization of a subject is given by

$$\frac{dA}{dt} = k_1(M - A) - k_2 A$$

Here $k_1 > 0,\ k_2 > 0$ and $A(t)$ is the amount of material memorized in time t, M is the total amount to be memorized and $M - A$ is the amount remaining to be memorized.

Lecture 13

Higher Order Linear Differential Equations

Preliminary theory

- A differential equation of the form

$$a_n(x)\frac{d^n y}{dx^n} + a_{n-1}(x)\frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x)\frac{dy}{dx} + a_0(x)y = g(x)$$

or $$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y = g(x)$$

where $a_0(x), a_1(x), \ldots, a_n(x), g(x)$ are functions of x and $a_n(x) \neq 0$, is called a linear differential equation with variable coefficients.

- However, we shall first study the differential equations with constant coefficients i.e. equations of the type

$$a_n\frac{d^n y}{dx^n} + a_{n-1}\frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1\frac{dy}{dx} + a_0 y = g(x)$$

where $a_0, a_1, ..., a_n$ are real constants. This equation is non-homogeneous differential equation and

- If $g(x) = 0$ then the differential equation becomes

$$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1 \frac{dy}{dx} + a_0 y = 0$$

Which is known as the associated homogeneous differential equation.

Initial -Value Problem

For a linear nth-order differential equation, the problem:

Solve: $$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x) y = g(x)$$

Subject to: $y(x_0) = y_0, \quad y'(x_0) = y_0', \ldots y^{n-1}(x_0) = y_0^{n-1}$

$y_0, y_0', ..., y_0^{n-1}$ being arbitrary constants, is called an initial-value problem (IVP).

The specified values $y(x_0) = y_0, y'(x_0) = y_0', ..., y^{n-1}(x_0) = y_0^{n-1}$ are called initial-conditions.

For $n = 2$ the initial-value problem reduces to

Solve: $$a_2(x) \frac{d^2 y}{dx^2} + a_1(x) \frac{dy}{dx} + a_0(x) y = g(x)$$

Subject to: $y(x_0) = y_0, \ldots, \quad y'(x_0) = y_0'$

Solution of IVP

A function satisfying the differential equation on *I* whose graph passes through (x_0, y_0) such that the slope of the curve at the point is the number y_0' is called solution of the initial value problem.

Boundary-value problem (BVP)

For a 2nd order linear differential equation, the problem

Solve: $$a_2(x)\frac{d^2y}{dx^2}+a_1(x)\frac{dy}{dx}+a_0(x)y=g(x)$$

Subject to: $y(a)=y_0,\quad y(b)=y_1$

is called a boundary-value problem. The specified values $y(a)=y_0$, and $y(b)=y_1$ are called boundary conditions.

Possible Boundary Conditions

For a 2nd order linear non-homogeneous differential equation

$$a_2(x)\frac{d^2y}{dx^2}+a_1(x)\frac{dy}{dx}+a_0(x)y=g(x)$$

all the possible pairs of boundary conditions are

$$y(a)=y_0,\qquad y(b)=y_1,$$

$$y'(a)=y'_0,\qquad y(b)=y_1,$$

$$y(a)=y_0,\qquad y'(b)=y'_1,$$

$$y'(a)=y'_0,\qquad y'(b)=y'_1$$

where y_0, y'_0, y_1 and y'_1 denote the arbitrary constants.

In General

All the four pairs of conditions mentioned above are just special cases of the general boundary conditions

$$\alpha_1 y(a)+\beta_1 y'(a)=\gamma_1$$
$$\alpha_2 y(b)+\beta_2 y'(b)=\gamma_2$$

where $\alpha_1, \alpha_2, \beta_1, \beta_2 \in \{0,1\}$

Note that

A boundary value problem may have

- Several solutions.
- A unique solution, or
- No solution at all.

Definition: Linear Dependence

A set of functions

$$\{f_1(x), f_2(x), \ldots, f_n(x)\}$$

is said to be *linearly dependent* on an interval *I* if $\exists$ constants $c_1, c_2, \ldots, c_n$ not all zero, such that

$$c_1 f_1(x) + c_2 f_2(x) + \cdots + c_n f_n(x) = 0, \quad \forall\, x \in I$$

Definition: Linear Independence

A set of functions

$$\{f_1(x), f_2(x), \ldots, f_n(x)\}$$

is said to be linearly independent on an interval *I* if

$$c_1 f_1(x) + c_2 f_2(x) + \cdots + c_n f_n(x) = 0, \quad \forall\, x \in I,$$

only when

$$c_1 = c_2 = \cdots = c_n = 0.$$

Case of two functions:

If $n = 2$ then the set of functions becomes

$$\{f_1(x), f_2(x)\}$$

If we suppose that

$$c_1 f_1(x) + c_2 f_2(x) = 0$$

Also that the functions are linearly dependent on an interval I then either $c_1 \neq 0$ or $c_2 \neq 0$.

Let us assume that $c_1 \neq 0$, then

$$f_1(x) = -\frac{c_2}{c_1} f_2(x);$$

Hence $f_1(x)$ is the constant multiple of $f_2(x)$.

Conversely, if we suppose

$$f_1(x) = c_2\, f_2(x)$$

Then $$(-1) f_1(x) + c_2 f_2(x) = 0, \quad \forall x \in I$$

So that the functions are linearly dependent because $c_1 = -1$.

Definition: Wronskian

Suppose that the function $f_1(x), f_2(x), \ldots, f_n(x)$ possesses at least $n-1$ derivatives then the determinant

$$\begin{vmatrix} f_1 & f_2 & \cdots\cdots & f_n \\ f_1' & f_2' & \cdots\cdots & f_n' \\ \vdots & \vdots & & \vdots \\ f_1^{n-1} & f_2^{n-1} & \cdots\cdots & f_n^{n-1} \end{vmatrix}$$

is called Wronskian of the functions $f_1(x), f_2(x), \ldots, f_n(x)$ and is denoted by $W\left(f_1(x), f_1(x), \ldots, f_1(x)\right)$.

Lecture 14

Solutions of Higher Order Linear Equations

Preliminary Theory

- In order to solve an *n*th order non-homogeneous linear differential equation

$$a_n(x)\frac{d^n y}{dx^n}+a_{n-1}(x)\frac{d^{n-1}y}{dx^{n-1}}+\cdots+a_1(x)\frac{dy}{dx}+a_0(x)y=g(x)$$

we first solve the associated homogeneous differential equation

$$a_n(x)\frac{d^n y}{dx^n}+a_{n-1}(x)\frac{d^{n-1}y}{dx^{n-1}}+\cdots+a_1(x)\frac{dy}{dx}+a_0(x)y=0$$

- Therefore, we first concentrate upon the preliminary theory and the methods of solving the homogeneous linear differential equation.

- We recall that a function $y=f(x)$ that satisfies the associated homogeneous equation

$$a_n(x)\frac{d^n y}{dx^n}+a_{n-1}(x)\frac{d^{n-1}y}{dx^{n-1}}+\cdots+a_1(x)\frac{dy}{dx}+a_0(x)y=0$$

Is called solution of the differential equation.

Superposition Principle

- Suppose that $y_1, y_2, ---, y_n$ are solutions on an interval I of the homogeneous linear differential equation

$$a_n(x)\frac{d^n y}{dx^n}+a_{n-1}(x)\frac{d^{n-1}y}{dx^{n-1}}+\cdots+a_1(x)\frac{dy}{dx}+a_0(x)y=0$$

Then

$y=c_1y_1(x)+c_2y_2(x)+\cdots+c_ny_n(x),$

- $c_1, c_2, \ldots, c_n$ being arbitrary constants is also a solution of the differential equation.

- A constant multiple $y=c_1y_1(x)$ of a solution $y_1(x)$ of the homogeneous linear differential equation is also a solution of the equation.
- The homogeneous linear differential equations always possess the trivial solution $y=0$.
- The superposition principle is a property of linear differential equations and it does not hold in case of non-linear differential equations.

Eample:

The function

$$y=x^2$$

Is a solution of the homogeneous linear equation?

$$x^2y''-3xy'+4y=0$$

on $(0,\infty)$.

Now consider

$$y=cx^2$$

Then $y' = 2cx$ and $y'' = 2c$

So that $x^2y'' - 3xy' + 4y = 2cx^2 - 6cx^2 + 4cx^2 = 0$

Hence the function

$$y = cx^2$$

is also a solution of the given differential equation.

The Wronskian

Suppose that y_1, y_2 are 2 solutions, on an interval I, of the second order homogeneous linear differential equation

$$a_2\frac{d^2y}{dx^2} + a_1\frac{dy}{dx} + a_0y = 0$$

Then either $W(y_1, y_2) = 0, \quad \forall\, x \in I$

or $W(y_1, y_2) \neq 0, \quad \forall\, x \in I$

To verify this we write the equation as

$$\frac{d^2y}{dx^2} + \frac{Pdy}{dx} + Qy = 0$$

Now $W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1y_2' - y_1'y_2$

Differentiating *w.r.to* x, we have

$$\frac{dW}{dx} = y_1y_2'' - y_1''y_2$$

Since y_1 and y_2 are solutions of the differential equation

$$\frac{d^2y}{dx^2} + \frac{Pdy}{dx} + Qy = 0$$

Therefore

$$y_1'' + Py_1' + Qy_1 = 0$$

$$y_2'' + Py_2' + Qy_2 = 0$$

Multiplying 1st equation by y_2 and 2nd by y_1 the have

$$y_1''y_2 + Py_1'y_2 + Qy_1y_2 = 0$$

$$y_1y_2'' + Py_1y_2' + Qy_1y_2 = 0$$

Subtracting the two equations we have:

$$(y_1y_2'' - y_2y_1'') + P(y_1y_2' - y_1'y_2) = 0$$

or $$\frac{dW}{dx} + PW = 0$$

This is a linear 1st order differential equation in W, whose solution is

$$W = ce^{-\int Pdx}$$

Therefore

q If $c \neq 0$ then $W(y_1, y_2) \neq 0, \quad \forall\, x \in I$

q If $c = 0$ then $W(y_1, y_2) = 0, \quad \forall\, x \in I$

Hence Wronskian of y_1 and y_2 is either identically zero or is never zero on I.

In general

If $y_1, y_2, \ldots, y_n$ are n solutions, on an interval I, of the homogeneous *n*th order linear differential equation with constants coefficients

$$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1 \frac{dy}{dx} + a_0 y = 0$$

Then

Either $W(y_1, y_2, \ldots, y_n) = 0, \quad \forall x \in I$

or $W(y_1, y_2, \ldots, y_n) \neq 0, \quad \forall x \in I$

Linear Independence of Solutions:

Suppose that

$$\boldsymbol{y}_1, \boldsymbol{y}_2, \ldots, \boldsymbol{y}_n$$

are *n* solutions, on an interval I, of the homogeneous linear *n*th-order differential equation

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x) y = 0$$

Then the set of solutions is linearly independent on *I* if and only if

$$W(y_1, y_2, \ldots, y_n) \neq 0$$

In other words

The solutions

$$y_1, y_2, \ldots, y_n$$

are linearly dependent if and only if

$$W(\ldots, y_n) = 0, \quad \forall x \in I$$

Fundamental Set of Solutions

A set

$$\{y_1, y_2, \ldots, y_n\}$$

of *n* linearly independent solutions, on interval I, of the homogeneous linear *n*th-order differential equation

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x) y = 0$$

is said to be a fundamental set of solutions on the interval I.

Existence of a Fundamental Set

There always exists a fundamental set of solutions for a linear *n*th-order homogeneous differential equation

$$a_n(x)\frac{d^n y}{dx^n}+a_{n-1}(x)\frac{d^{n-1}y}{dx^{n-1}}+\cdots+a_1(x)\frac{dy}{dx}+a_0(x)y=0$$

on an interval *I.*

Eample:

The functions

$$y_1=e^{3x} \text{ and } y_2=e^{-3x}$$

are solutions of the differential equation

$$y''-9y=0$$

Since $W\left(e^{3x},e^{-3x}\right)=\begin{vmatrix} e^{3x} & e^{-3x} \\ 3e^{3x} & -3e^{-3x} \end{vmatrix}=-6\neq 0, \quad \forall x\in I$

Therefore y_1 and y_2 from a fundamental set of solutions on $(-\infty,\infty)$. **Hence** general solution of the differential equation on the $(-\infty,\infty)$ is

$$y=c_1e^{3x}+c_2e^{-3x}$$

Eample:

Consider the differential equation

$$\frac{d^3y}{dx^3}-6\frac{d^2y}{dx^2}+11\frac{dy}{dx}-6y=0$$

and suppose that $y_1=e^x,\ y_2=e^{2x}$ and $y_3=e^{3x}$

Then $\frac{dy_1}{dx}=e^x=\frac{d^2y_1}{dx^2}=\frac{d^3y_1}{dx^3}$

Therefore $\frac{d^3y_1}{dx^3}-6\frac{d^2y_1}{dx^2}+11\frac{dy_1}{dx}-6y_1=e^x-6e^x+11e^x-6e^x$

or $\frac{d^3y_1}{dx^3}-6\frac{d^2y_1}{dx^2}+11\frac{dy_1}{dx}-6y_1=12e^x-12e^x=0$

Thus the function y_1 is a solution of the differential equation. Similarly, we can verify that the other two functions i.e. y_2 and y_3 also satisfy the differential equation.

Now for all $x\in R$

$$W\left(e^x,e^{2x},e^{3x}\right)=\begin{vmatrix} e^x & e^{2x} & e^{3x} \\ e^x & 2e^{2x} & 3e^{3x} \\ e^x & 4e^{2x} & 9e^{3x} \end{vmatrix}=2e^{6x}\neq 0 \quad \forall x\in I$$

Therefore $y_1,y_2,$ and y_3 form a fundamental solution of the differential equation on $(-\infty,\infty)$. We conclude that

$$y=c_1e^x+c_2e^{2x}+c_3e^{3x}$$

is the general solution of the differential equation on the interval $(-\infty,\infty)$.

Non-Homogeneous Equations

A function y_p that satisfies the non-homogeneous differential equation

$$a_n(x)\frac{d^n y}{dx^n}+a_{n-1}(x)\frac{d^{n-1} y}{dx^{n-1}}+\cdots+a_1(x)\frac{dy}{dx}+a_0(x)y=g(x)$$ And is free of parameters is called the particular solution of the differential equation.

Eample:

Suppose that

$$y_p = 3$$

Then $$y_p'' = 0$$

So that

$$y_p'' + 9y_p = 0 + 9(3)$$
$$= 27$$

Therefore

$$y_p = 3$$

is a particular solution of the differential equation

$$y_p'' + 9y_p = 27$$

Complementary Function

The general solution

$$y_c = c_1 y_1 + c_2 y_2 + \cdots + c_n y_n$$

of the homogeneous linear differential equation

$$a_n(x)\frac{d^n y}{dx^n}+a_{n-1}(x)\frac{d^{n-1} y}{dx^{n-1}}+\cdots+a_1(x)\frac{dy}{dx}+a_0(x)y=0$$

is known as the complementary function for the non-homogeneous linear differential equation.

$$a_n(x)\frac{d^n y}{dx^n}+a_{n-1}(x)\frac{d^{n-1} y}{dx^{n-1}}+\cdots+a_1(x)\frac{dy}{dx}+a_0(x)y=g(x)$$

General Solution of Non-Homogeneous Equations

Suppose that

q The particular solution of the non-homogeneous equation

$$a_n(x)\frac{d^n y}{dx^n}+a_{n-1}(x)\frac{d^{n-1} y}{dx^{n-1}}+\cdots+a_1(x)\frac{dy}{dx}+a_0(x)y=g(x)$$

is y_p.

The complementary function of the non-homogeneous differential equation

$$a_n(x)\frac{d^n y}{dx^n}+a_{n-1}(x)\frac{d^{n-1} y}{dx^{n-1}}+\cdots+a_1(x)\frac{dy}{dx}+a_0(x)y=0$$

is

$$y_c = c_1 y_1 + c_2 y_2 + \cdots + c_n y_n.$$

Then general solution of the non-homogeneous equation on the interval I is given by

$$y = y_c + y_p$$

or

$$y = c_1 y_1(x) + c_2 y_2(x) + \cdots + c_n y_n(x) + y_p(x) = y_c(x) + y_p(x)$$

Hence

General Solution = Complementary solution + any particular solution.

Superposition Principle for Non-homogeneous Equations

Suppose that

$$y_{p_1}, y_{p_2}, \ldots, y_{p_k}$$

denote the particular solutions of the k differential equation

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y = g_i(x),$$

$i = 1,2,\ldots k$, on an interval I. Then

$$y_p = y_{p_1}(x) + y_{p_2}(x) + \cdots + y_{p_k}(x)$$

is a particular solution of

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y = g_1(x) + g_2(x) + \cdots + g_k(x)$$

Eample:

Consider the differential equation

$$y'' - 3y' + 4y = -16x^2 + 24x - 8 + 2e^{2x} + 2xe^x - e^x$$

Suppose that

$$y_{p_1} = -4x^2, \qquad y_{p_2} = e^{2x}, \qquad y_{p_3} = xe^x$$

Then $\quad y''_{p_1} - 3y'_{p_1} + 4y_{p_1} = -8 + 24x - 16x^2$

Therefore $\quad y_{p_1} = -4x^2$

is a particular solution of the non-homogenous differential equation

$$y'' - 3y' + 4y = -16x^2 + 24x - 8$$

Similarly, it can be verified that

$$y_{p_2} = e^{2x} \quad \text{and} \quad y_{p_3} = xe^x$$

are particular solutions of the equations:

$$y'' - 3y' + 4y = 2e^{2x}$$

and $\quad y'' - 3y' + 4y = 2xe^x - e^x$

respectively.

Hence $\quad y_p = y_{p_1} + y_{p_2} + y_{p_3} = -4x^2 + e^{2x} + xe^x$

is a particular solution of the differential equation

$$y'' - 3y' + 4y = -16x^2 + 24x - 8 + 2e^{2x} + 2xe^x - e^x$$

Lecture 15

Construction of a Second Solution

Construction of a Second Solution General Case

Consider the differential equation

$$a_2(x)\frac{d^2y}{dx^2}+a_1(x)\frac{dy}{dx}+a_0(x)y=0$$

We divide by $a_2(x)$ to put the above equation in the form

$$y''+P(x)y'+Q(x)y=0$$

Where $P(x)$ and $Q(x)$ are continuous on some interval I.

Suppose that $y_1(x)\neq 0,\ \forall\ x\in \mathrm{I}$ is a solution of the differential equation

Then $$y_1''+P\,y_1'+Q\,y_1=0$$

We define $y=u(x)y_1(x)$ then

$$y'=uy_1'+y_1u' \qquad y''=uy_1''+2y_1'u_1'+y_1u''$$

$$y''+Py'+Qy=u[\underbrace{y_1''+Py_1'+Qy_1}_{zero}]+y_1u''+(2y_1'+Py_1)u'=0$$

This implies that we must have

$$y_1u''+(2y_1'+Py_1)u'=0$$

If we suppose $w=u'$, then

$$y_1w'+(2y_1'+Py_1)w=0$$

The equation is separable. Separating variables we have from the last equation

$$\frac{dw}{w}+(2\frac{y_1'}{y_1}+P)dx=0$$

.

Integrating
$$\ln|w|+2\ln|y_1|=-\int Pdx+c$$
$$\ln|wy_1^2|=-\int Pdx+c$$
$$wy_1^2=c_1e^{-\int Pdx}$$
$$w=\frac{c_1e^{-\int Pdx}dx}{y_1^2}$$
$$u'=\frac{c_1e^{-\int Pdx}}{y_1^2}$$

or

Integrating again, we obtain

$$u=c_1\int\frac{e^{-\int Pdx}}{y_1^2}dx+c_2$$

$$y=u(x)y_1(x)=c_1y_1(x)\int\frac{e^{-\int Pdx}}{y_1^2}dx+c_2y_1(x).$$

Hence

Choosing $c_1 = 1$ and $c_2 = 0$, we obtain a second solution of the differential equation

$$y_2 = y_1(x)\int \frac{e^{-\int P dx}}{{y_1}^2}dx$$

The Wronskian

$$W(y_1(x), y_2(x)) = \begin{vmatrix} y_1 & y_1\int \frac{e^{-\int P dx}}{{y_1}^2}dx \\ y_1' & \frac{e^{-\int P dx}}{y_1} + y_1'\int \frac{e^{-\int P dx}}{{y_1}^2}dx \end{vmatrix}$$

$$= e^{-\int P dx} \neq \mathbf{0}, \forall x$$

Therefore $y_1(x)$ and $y_2(x)$ are linear independent set of solutions. So that they form a fundamental set of solutions of the differential equation

$$y'' + P(x)y' + Q(x)y = 0$$

Hence the general solution of the differential equation is
$y(x) = c_1 y_1(x) + c_2 y_2(x)$

Example

Given that

$$y_1 = x^2$$

is a solution of

$$x^2 y'' - 3xy' + 4y = 0$$

Find general solution of the differential equation on the interval $(0, \infty)$.

Solution:

The equation can be written as

$$y'' - \frac{3}{x}y' + \frac{4}{x^2}y = 0,$$

The 2nd solution y_2 is given by

$$y_2 = y_1(x)\int \frac{e^{-\int P dx}}{{y_1}^2}dx$$

or

$$y_2 = x^2\int \frac{e^{3\int dx/x}}{x^4}dx = x^2\int \frac{e^{\ln x^3}}{x^4}dx$$

$$y_2 = x^2\int \frac{1}{x}dx = x^2 \ln x$$

Hence the general solution of the differential equation on $(0, \infty)$ is given by

$$y = c_1 y_1 + c_2 y_2$$

or

$$y = c_1 x^2 + c_2 x^2 \ln x$$

Lecture 16

Homogeneous Linear Equations with Constant Coefficients

Homogeneous Linear Equations with Constant Coefficients

We know that the linear first order differential equation

$$\frac{dy}{dx}+my=0$$

m being a constant, has the exponential solution on $(-\infty,\infty)$

$$y=c_1e^{-mx}$$

The question

The question is whether or not the exponential solutions of the higher-order differential equations

$$a_ny^{(n)}+a_{n-1}y^{(n-1)}+\cdots+a_2y^{//}+a_1y^{/}+a_0y=0,$$

exist on $(-\infty,\infty)$.

In fact all the solutions of this equation are exponential functions or constructed out of exponential functions.

Recall

That the linear differential of order n is an equation of the form

$$a_n(x)\frac{d^ny}{dx^n}+a_{n-1}(x)\frac{d^{n-1}y}{dx^{n-1}}+\cdots+a_1(x)\frac{dy}{dx}+a_0(x)y=g(x)$$

Method of Solution

Taking $n=2$, the *n*th-order differential equation becomes

$$a_2\frac{d^2y}{dx^2}+a_1\frac{dy}{dx}+a_0y=0$$

This equation can be written as

$$a\frac{d^2y}{dx^2}+b\frac{dy}{dx}+cy=0$$

We now try a solution of the exponential form

$$y=e^{mx}$$

Then

$$y'=me^{mx} \text{ and } y''=m^2e^{mx}$$

Substituting in the differential equation, we have

$$e^{mx}(am^2+bm+c)=0$$

Since $\quad e^{mx}\neq 0, \quad \forall x\in(-\infty,\infty)$

Therefore $\quad am^2+bm+c=0$

This algebraic equation is known as the Auxiliary equation (AE).The solution of the auxiliary equation determines the solutions of the differential equation.

Case 1: *Distinct Real Roots*

If the auxiliary equation has distinct real roots m_1 and m_2 then we have the following two solutions of the differential equation.

$$y_1 = e^{m_1 x} \text{ and } y_2 = e^{m_2 x}$$

These solutions are linearly independent because

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = (m_2 - m_1)e^{(m_1+m_2)x}$$

Since $m_1 \neq m_2$ and $e^{(m_1+m_2)x} \neq 0$

Therefore $W(y_1, y_2) \neq 0 \;\; \forall x \in (-\infty, \infty)$

Hence

y_1 and y_2 form a fundamental set of solutions of the differential equation.

The general solution of the differential equation on $(-\infty, \infty)$ is

$$y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$$

Case 2. *Repeated Roots*

If the auxiliary equation has real and equal roots i.e

$$m = m_1, m_2 \quad \text{with} \quad m_1 = m_2$$

Then we obtain only one exponential solution

$$y = c_1 e^{mx}$$

To construct a second solution we rewrite the equation in the form

$$y'' + \frac{b}{a} y' + \frac{c}{a} y = 0$$

Comparing with $y'' + Py' + Qy = 0$

We make the identification

$$P = \frac{b}{a}$$

Thus a second solution is given by

$$y_2 = y_1 \int \frac{e^{-\int P dx}}{y_1^2} dx = e^{mx} \int \frac{e^{-\frac{b}{a}x}}{e^{2mx}} dx$$

Since the auxiliary equation is a quadratic algebraic equation and has equal roots

Therefore, $Disc. = b^2 - 4ac = 0$

We know from the quadratic formula

$$m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

we have $2m = -\frac{b}{a}$

Therefore

$$y_2 = e^{mx}\int \frac{e^{2mx}}{e^{2mx}}dx = xe^{mx}$$

Hence the general solution is

$$y = c_1 e^{mx} + c_2 x e^{mx} = (c_1 + c_2 x)e^{mx}$$

Case 3**:** *Complex Roots*

If the auxiliary equation has complex roots $\alpha \pm i\beta$ then, with

$m_1 = \alpha + i\beta$ and $m_2 = \alpha - i\beta$

Where α>0 and β>0 are real, the general solution of the differential equation is

$$y = c_1 e^{(\alpha+i\beta)x} + c_2 e^{(\alpha-i\beta)x}$$

First we choose the following two pairs of values of c_1 and c_2

$$c_1 = c_2 = 1$$

$$c_1 = 1, c_2 = -1$$

Then we have

$y_1 = e^{(\alpha+i\beta)x} + e^{(\alpha-i\beta)x}$

$y_2 = e^{(\alpha+i\beta)x} - e^{(\alpha-i\beta)x}$

We know by the Euler's Formula that

$$e^{i\theta} = \cos\theta + i\sin\theta, \quad \theta \in \mathrm{R}$$

Using this formula, we can simplify the solutions y_1 and y_2 as

$$y_1 = e^{\alpha x}(e^{i\beta x} + e^{-i\beta x}) = 2e^{\alpha x}\cos\beta x$$

$$y_2 = e^{\alpha x}(e^{i\beta x} - e^{-i\beta x}) = 2ie^{\alpha x}\sin\beta x$$

We can drop constant to write

$$y_1 = e^{\alpha x}\cos\beta x, \quad y_2 = e^{\alpha x}\sin\beta x$$

The Wronskian

$W\left(e^{\alpha x}\cos\beta x, e^{\alpha x}\sin\beta x\right) = \beta e^{2\alpha x} \neq 0 \quad \forall\ x$

Therefore, $e^{\alpha x}\cos(\beta x),\ e^{\alpha x}\sin(\beta x)$

form a fundamental set of solutions of the differential equation on $(-\infty,\infty)$.
Hence general solution of the differential equation is

$$y = c_1 e^{\alpha x}\cos\beta x + c_2 e^{\alpha x}\sin\beta x$$

or

$$y = e^{\alpha x}(c_1\cos\beta x + c_2\sin\beta x)$$

Higher Order Equations

If we consider nth order homogeneous linear differential equation

$$a_n\frac{d^n y}{dx^n} + a_{n-1}\frac{d^{n-1}y}{dx^{n-1}} + \ldots + a_1\frac{dy}{dx} + a_0 y = 0$$

Then, the auxiliary equation is an nth degree polynomial equation

$$a_n m^n + a_{n-1}m^{n-1} + \ldots + a_1 m + a_0 = 0$$

Case 1: Real distinct roots

If the roots $m_1, m_2, \ldots, m_n$ of the auxiliary equation are all real and distinct, then the general solution of the equation is

$$y = c_1 e^{m_1 x} + c_2 e^{m_2 x} + \ldots + c_n e^{m_n x}$$

Case 2: Real & repeated roots

We suppose that m_1 is a root of multiplicity n of the auxiliary equation, then it can be shown that

$$e^{m_1 x}, xe^{m_1 x}, \ldots, x^{n-1}e^{m_1 x}$$

are n linearly independent solutions of the differential equation. Hence general solution of the differential equation is

$$y = c_1 e^{m_1 x} + c_2 xe^{m_1 x} + \ldots + c_n x^{n-1}e^{m_1 x}$$

Case 3: Complex roots

- Suppose that coefficients of the auxiliary equation are real.
  We fix n at 6, all roots of the auxiliary are complex,
  $y = e^{\alpha x}\left[\left(c_1 + c_2 x + c_3 x^2\right)\cos\beta x + \left(d_1 + d_2 x + d_3 x^2\right)\sin\beta x\right]$
  namely $\alpha_1 \pm i\beta_1,\quad \alpha_2 \pm i\beta_2,\ \alpha_3 \pm i\beta_3$
- Then the general solution of the differential equation
  $y = e^{\alpha_1 x}(c_1\cos\beta_1 x + c_2\sin\beta_1 x) + e^{\alpha_2 x}(c_3\cos\beta_2 x + c_4\sin\beta_2 x)$
  $+ e^{\alpha_3 x}(c_5\cos\beta_3 x + c_6\sin\beta_3 x)$
- If $n = 6$, two roots of the auxiliary equation are real and equal and the remaining 4 are complex, namely $\alpha_1 \pm i\beta_1,\quad \alpha_2 \pm i\beta_2$

Then the general solution is

$$y = e^{\alpha_1 x}(c_1 \cos \beta_1 x + c_2 \sin \beta_1 x) + e^{\alpha_2 x}(c_3 \cos \beta_2 x + c_4 \sin \beta_2 x) + c_5 e^{m_1 x} + c_6 x e^{m_1 x}$$

- If $m_1 = \alpha + i\beta$ is a complex root of multiplicity k of the auxiliary equation. Then its conjugate $m_2 = \alpha - i\beta$ is also a root of multiplicity k. Thus from Case 2 **,** the differential equation has $2k$ solutions

$$e^{(\alpha+i\beta)x},\ xe^{(\alpha+i\beta)x},\ x^2 e^{(\alpha+i\beta)x},\ldots, x^{k-1}e^{(\alpha+i\beta)x}$$

$$e^{(\alpha-i\beta)x},\ xe^{(\alpha-i\beta)x},\ x^2 e^{(\alpha-i\beta)x},\ldots, x^{k-1}e^{(\alpha-i\beta)x}$$

- By using the Euler's formula, we conclude that the general solution of the differential equation is a linear combination of the linearly independent solutions

$$e^{\alpha x}\cos \beta x,\ xe^{\alpha x}\cos \beta x,\ x^2 e^{\alpha x}\cos \beta x,\ldots, x^{k-1}e^{\alpha x}\cos \beta x$$

$$e^{\alpha x}\sin \beta x,\ xe^{\alpha x}\sin \beta x,\ x^2 e^{\alpha x}\sin \beta x,\ldots, x^{k-1}e^{\alpha x}\sin \beta x$$

- Thus if $k = 3$ then

- $y = e^{\alpha x}\left[\left(c_1 + c_2 x + c_3 x^2\right)\cos \beta x + \left(d_1 + d_2 x + d_3 x^2\right)\sin \beta x\right]$

Solving the Auxiliary Equation

- Recall that the auxiliary equation of nth degree differential equation is nth degree polynomial equation
- Solving the auxiliary equation could be difficult

$$P_n(m) = 0, \quad n > 2$$

- One way to solve this polynomial equation is to guess a root m_1. Then $m - m_1$ is a factor of the polynomial $P_n(m)$.
- Dividing with $m - m_1$ synthetically or otherwise, we find the factorization $P_n(m) = (m - m_1)\,Q(m)$
- We then try to find roots of the quotient i.e. roots of the polynomial equation

$$Q(m) = 0$$

- Note that if $m_1 = \dfrac{p}{q}$ is a rational real root of the equation

$$P_n(m) = 0, \quad n > 2$$

then p is a factor of a_0 and q of a_n.

- By using this fact we can construct a list of all possible rational roots of the auxiliary equation and test each of them by synthetic division.

Lecture 17

Method of Undetermined Coefficients-Superposition Approach

Recall

1. That a non-homogeneous linear differential equation of order n is an equation of the form

$$a_n \frac{d^n y}{dx^n} + a_{n-1}\frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1 \frac{dy}{dx} + a_0 y = g(x)$$

The coefficients $a_0, a_1, \ldots, a_n$ can be functions of x. However, we will discuss equations with constant coefficients.

2. That to obtain the general solution of a non-homogeneous linear differential equation we must find**:**

The complementary function y_c, which is general solution of the associated homogeneous differential equation.

Any particular solution y_p of the non-homogeneous differential equation.

3. That the general solution of the non-homogeneous linear differential equation is given by *General solution = Complementary function + Particular Integral*

Finding

Complementary function has been discussed in the previous lecture. In the next three lectures we will discuss methods for finding a particular integral for the non-homogeneous equation, namely

- The method of undetermined coefficients-*superposition approach*
- The method undetermined coefficients-*annihilator operator approach.*
- The method of variation of parameters.

The Method of Undetermined Coefficient

The method of undetermined coefficients developed here is limited to non-homogeneous linear differential equations

- That have constant coefficients, and
- Where the function $g(x)$ has a specific form.

The form of $g(x)$

The input function $g(x)$ can have one of the following forms:

- A constant function *k*.
- A polynomial function
- An exponential function e^x
- The trigonometric functions $\sin(\beta x)$, $\cos(\beta x)$
- Finite sums and products of these functions.
- *Otherwise*, we cannot apply the method of undetermined coefficients.

The method

Consist of performing the following steps.

Step 1 Determine the form of the input function $g(x)$.

Step 2 Assume the general form of y_p according to the form of $g(x)$
Step 3 Substitute in the given non-homogeneous differential equation.
Step 4 Simplify and equate coefficients of like terms from both sides.
Step 5 Solve the resulting equations to find the unknown coefficients.

Step 6 Substitute the calculated values of coefficients in assumed y_p

Restriction on ?

The input function g is restricted to have one of the above stated forms because of the reason:

- The derivatives of sums and products of polynomials, exponentials etc are again sums and products of similar kind of functions.
- The expression $ay_p'' + by_p' + cy_p$ has to be identically equal to the input function $g(x)$.

  Therefore, to make an educated guess, y_p is assured to have the same form as g.

Trial particular solutions

Number	The input function $g(x)$	The assumed particular solution y_p
1	Any constant e.g. 1	A
2	$5x+7$	$Ax+B$
3	$3x^2-2$	Ax^2+Bx+c
4	x^3-x+1	Ax^3+Bx^2+Cx+D
5	$\sin 4x$	$A\cos 4x+B\sin 4x$
6	$\cos 4x$	$A\cos 4x+B\sin 4x$
7	e^{5x}	Ae^{5x}
8	$(9x-2)e^{5x}$	$(Ax+B)e^{5x}$
9	x^2e^{5x}	$(Ax^2+Bx+C)e^{5x}$
10	$e^{3x}\sin 4x$	$Ae^{3x}\cos 4x+Be^{3x}\sin 4x$
11	$5x^2\sin 4x$	$(A_1x^2+B_1x+C_1)\cos 4x+(A_2x^2+B_2x+C_2)\sin 4x$
12	$xe^{3x}\cos 4x$	$(Ax+B)e^{3x}\cos 4x+(Cx+D)e^{3x}\sin 4x$

If $g(x)$ equals a sum?

Suppose that

The input function $g(x)$ consists of a sum of m terms of the kind listed in the above table i.e.

$$g(x)=g_1(x)+g_2(x)+\cdots+g_m(x)$$

The trial forms corresponding to $g_1(x), g_2(x), \ldots, g_m(x)$ be $y_{p_1}, y_{p_2}, \ldots, y_{p_m}$.

Then the particular solution of the given non-homogeneous differential equation is

$$y_p = y_{p_1} + y_{p_2} + \cdots + y_{p_m}$$

In other words, *the form of* y_p *is a linear combination of all the linearly independent functions generated by repeated differentiation of the input function* $g(x)$ *.*

Example:

Example 4

Find a particular solution of the following non-homogeneous differential equation

$$y'' - 5y' + 4y = 8e^x$$

Solution:

To find y_c, we solve the associated homogeneous differential equation

$$y'' - 5y' + 4y = 0$$

We put $y = e^{mx}$ in the given equation, so that the auxiliary equation is

$$m^2 - 5m + 4 = 0 \Rightarrow m = 1, 4$$

Thus $\quad y_c = c_1e^x + c_2e^{4x}$

Since $\quad g(x) = 8e^x$

Therefore, $\quad y_p = Ae^x$

Substituting in the given non-homogeneous differential equation, we obtain

$$Ae^x - 5Ae^x + 4Ae^x = 8e^x$$

So $\quad 0 = 8e^x$

Clearly we have made a wrong assumption for y_p, as we did not remove the duplication.

Since Ae^x is present in y_c. Therefore, it is a solution of the associated homogeneous differential equation

$$y'' - 5y' + 4y = 0$$

To avoid this we find a particular solution of the form

$$y_p = Axe^x$$

We notice that there is no duplication between y_c and this new assumption for y_p

Now $\quad y_p' = Axe^x + Ae^x, \quad y_p'' = Axe^x + 2Ae^x$

Substituting in the given differential equation, we obtain

$$Axe^x + 2Ae^x - 5Axe^x - 5Ae^x + 4Axe^x = 8e^x.$$

or $\quad -3Ae^x = 8e^x \Rightarrow A = -8/3.$

So that a particular solution of the given equation is given by

$$y_p = -(8/3)e^x$$

Hence, the general solution of the given equation is

$$y = c_1e^x + c_2e^{4x} - (8/3)\ x\ e^x$$

Example:

Example 12
Determine the form of a particular solution of the equation

$$y'''' + y''' = 1 - e^{-x}$$

Solution:
Consider the associated homogeneous differential equation

$$y'''' + y''' = 0$$

The auxiliary equation is

$$m^4 + m^3 = 0 \Rightarrow m = 0, 0, 0, -1$$

Therefore, the complementary function is

$$y_c = c_1 + c_2 x + c_3 x^2 + c_4 e^{-x}$$

Since $\quad g(x) = 1 - e^{-x} = g_1(x) + g_2(x)$

Corresponding to $g_1(x) = 1$: $\quad y_{p_1} = A$

Corresponding to $g_2(x) = -e^{-x}$: $\quad y_{p_2} = Be^{-x}$

Therefore, the normal assumption for the particular solution is

$$y_p = A + Be^{-x}$$

Clearly there is duplication of

(i) The constant function between y_c and y_{p_1}.

(ii) The exponential function e^{-x} between y_c and y_{p_2}.

To remove this duplication, we multiply y_{p_1} with x^3 and y_{p_2} with x. This duplication can't be removed by multiplying with x and x^2. Hence, the correct assumption for the particular solution y_p is

$$y_p = Ax^3 + Bxe^{-x}$$

Lecture 18

Undetermined Coefficient: Annihilator Operator Approach

Undetermined Coefficient: Annihilator Operator Approach

Recall

1. That a non-homogeneous linear differential equation of order n is an equation of the form

- $$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1 \frac{dy}{dx} + a_0 y = g(x)$$

- The following differential equation is called the associated homogeneous equation $$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1 \frac{dy}{dx} + a_0 y = 0$$
- The coefficients $a_0, a_1, \ldots, a_n$ can be functions of x. However, we will discuss equations with constant coefficients.

2. That to obtain the general solution of a non-homogeneous linear differential equation we must find:

- The complementary function y_c, which is general solution of the associated homogeneous differential equation.
- Any particular solution y_p of the non-homogeneous differential equation.

3. That the general solution of the non-homogeneous linear differential equation is given by

General Solution = Complementary Function + Particular Integral

- Finding the complementary function has been completely discussed in an earlier lecture
- In the previous lecture, we studied a method for finding particular integral of the non-homogeneous equations. This was the method of undetermined coefficients developed from the viewpoint of superposition principle.
- In the present lecture, we will learn to find particular integral of the non-homogeneous equations by the same method utilizing the concept of differential annihilator operators.

Differential Operators

- In calculus, the differential coefficient d/dx is often denoted by the capital letter D. So that $\frac{dy}{dx} = Dy$
- The symbol D is known as differential operator.

- This operator transforms a differentiable function into another function, e.g. $$D(e^{4x}) = 4e^{4x},\ D(5x^3 - 6x^2) = 15x^2 - 12x,\ D(\cos 2x) = -2\sin 2x$$
- The differential operator D possesses the property of linearity. This means that if f, g are two differentiable functions, then $$D\{af(x) + bg(x)\} = aDf(x) + bDg(x)$$

Where a and b are constants. Because of this property, we say that D is a linear differential operator.

- Higher order derivatives can be expressed in terms of the operator D in a natural manner:

$$\frac{d^2y}{d^2x} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = D(Dy) = D^2y$$

- Similarly

$$\frac{d^3y}{dx^3} = D^3y, \ldots, \frac{d^ny}{d^nx} = D^ny$$

- The following polynomial expression of degree n involving the operator D

- In calculus, the differential coefficient d/dx is often denoted by the capital letter D. So
-

$$a_nD^n + a_{n-1}D^{n-1} + \cdots + a_1D + a_0$$

is also a linear differential operator.

For example, the following expressions are all linear differential operators

$$D+3,\ D^2+3D-4,\ 5D^3-6D^2+4D$$

Differential Equation in Terms of *D*

Any linear differential equation can be expressed in terms of the notation D. Consider a 2nd order equation with constant coefficients

$$ay'' + by' + cy = g(x)$$

Since

$$\frac{dy}{dx} = Dy, \frac{d^2y}{dx^2} = D^2y$$

Therefore the equation can be written as

$$aD^2y + bDy + cy = g(x)$$

or

$$(aD^2 + bD + c)y = g(x)$$

Now, we define another differential operator L as

$$L = aD^2 + bD + c$$

Then the equation can be compactly written as

$$L(y) = g(x)$$

The operator L is a second-order linear differential operator with constant coefficients

Example:

Consider the differential equation

$$y'' + y' + 2y = 5x - 3$$

Since $\dfrac{dy}{dx} = Dy, \dfrac{d^2y}{dx^2} = D^2y$

Therefore, the equation can be written as

$$(D^2 + D + 2)y = 5x - 3$$

Now, we define the operator L as

$$L = D^2 + D + 2$$

Then the given differential can be compactly written as

$$L(y) = 5x - 3$$

Factorization of a differential operator

- An nth-order linear differential operator

  $$L = a_n D^n + a_{n-1} D^{n-1} + \cdots + a_1 D + a_0$$

  with constant coefficients can be factorized, whenever the characteristics polynomial equation

  $$L = a_n m^n + a_{n-1} m^{n-1} + \cdots + a_1 m + a_0$$

  can be factorized.

- The factors of a linear differential operator with constant coefficients commute.